TUTORIAL

Tutorial: structured models by the numbers

Four short problems to solve in class. Each problem is followed by its solution and step-by-step ‘explained’ slides.

1

Williams model

Steady-state chemostat: active fraction, residual glucose and yield vs D

Sect. 7.4.1, Ex. 7.5

2

LAB model

Closed-form steady states, maximum D and a growth threshold

Ex. 7.6

3

Wash-out memory

Why two cultures wash out at different rates after the same shift

Fig. 7.13

4

Cybernetic model

Resource allocation u_i, w_i, the switch point and the diauxic lag

Sect. 7.4.2

How to use: in Slide Show, hide the solution and explained slides (right-click → Hide Slide) and reveal them after students have worked the problem. A calculator is enough for Problems 1, 2 and 4(a–d); Problems 3 and 4(e) also have simple estimates students can check against the given exact answers.

Parameter values are illustrative, chosen to give realistic magnitudes; they are not the fitted values of the original papers. Answers were checked by numerical integration.

KEY IDEAS

Before you start: how to attack a chemostat problem

Problems 1 and 2 (and the start of 3) all follow the same four moves

The four moves

1 Biomass balance → μ = D

In a steady chemostat the cells must grow exactly as fast as they are washed out. So the dilution rate you set is the growth rate.

2 Internal balance → composition

Set d/dt = 0 for one cell component (X_G or X_A). This fixes how the cell is built (X_A, X_G) at that growth rate.

3 μ = D again → glucose

The growth expression contains glucose s, but only through the saturation factor f. Solve for f first, then convert f to s.

4 Everything else follows

Uptake rate v_1, yield $Y_{sx} = \mu/v_1$, limits ($s \to \infty$) and thresholds ($D \to 0$) are now plug-in calculations.

Why μ = D?

$$\frac{dx}{dt} = \underbrace{\mu x}_{\text{growth}} - \underbrace{Dx}_{\text{wash-out}} = 0$$

Sterile feed brings in no cells, so the only gain is growth and the only loss is outflow. With $x \neq 0$ this forces $\mu = D$.

Count before you solve

- Unknowns: X_A, X_G, s (three)
- Equations: $X_A + X_G = 1$, the X_G balance, $\mu = D$ (three)
- The skill is picking an order in which each step has only one new unknown.

x = biomass concentration (g DW L^{-1}); X_A, X_G = mass fractions of the active and structural/genetic compartments, so $X_A + X_G = 1$.

KEY IDEAS

What is f? The saturation factor

f (and f_1, f_2 in the LAB model) is just the glucose level rescaled to a 0-to-1 'how busy' number

Definition

$$f = \frac{s}{s + K_s}, \qquad v_1 = k_1 f$$

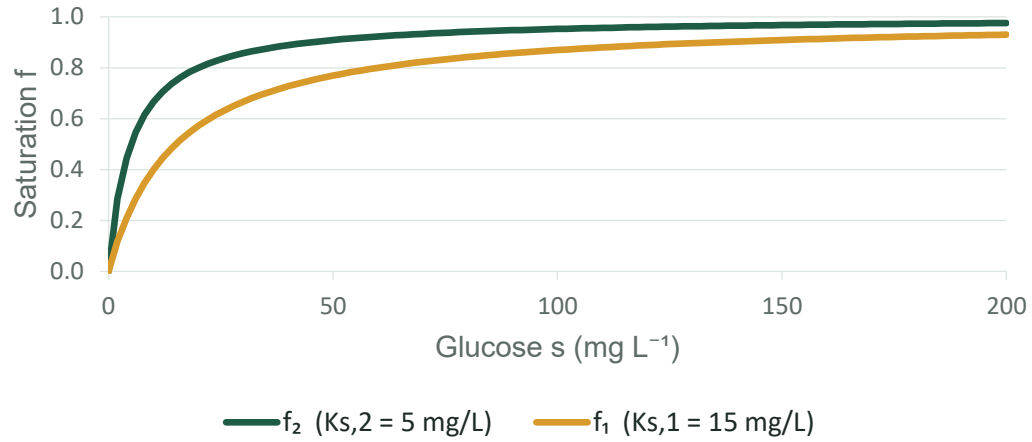
k_1 is the uptake rate if glucose were unlimited. f is the fraction of that maximum actually reached, so $v_1 = k_1 f$. K_s is the glucose level giving half speed.

Glucose s	0	K_s	$3K_s$	$9K_s$	$\gg K_s$
f	0	0.50	0.75	0.90	→ 1

Going back from f to s

$$f(s + K_s) = s \;\Rightarrow\; fK_s = s(1 - f) \;\Rightarrow\; s = K_s \frac{f}{1 - f}$$

Note how fast s grows as f → 1: f = 0.9 needs s = $9K_s$, f = 0.99 needs $99K_s$. f ≥ 1 is impossible — that is wash-out.



Why solve for f instead of s?

- s appears in the model only inside f, and μ is linear in f — so μ = D gives f with one division.
- Then $s = K_s\, f/(1 - f)$ converts back. Same answer as solving for s directly, much less algebra.

f_1 and f_2: the LAB model has two glucose-dependent reactions with different K_s values, so each gets its own saturation factor computed from the same s (Problem 2).

Problem 1 — Williams model in a chemostat

A culture following the Williams model (7.32a–c) runs at steady state in a glucose-limited chemostat with sterile feed

Given

Symbol	Value	Units
γ_{11}	0.50	g X_A (g S)$^{-1}$
γ_{22}	0.90	g X_G (g X_A)$^{-1}$
k_1	2.0	g S (g DW h)$^{-1}$
K_s	0.020	g L^{-1}
k_2	3.0	h^{-1}
k_3	0.10	h^{-1}

Model

$$v_1 = k_1 \frac{s}{s + K_s}, \quad v_2 = k_2 X_A X_G, \quad v_3 = k_3 X_G$$

$$\mu = \gamma_{11} v_1 - (1 - \gamma_{22}) v_2, \qquad X_A + X_G = 1$$

$$\frac{dX_G}{dt} = \gamma_{22} v_2 - v_3 - \mu X_G$$

Tasks

(a) Starting from the X_G balance, show that at steady state $X_A = (D + k_3) / (\gamma_{22} k_2)$.

(b) Compute X_A and X_G for D = 0.1, 0.3 and 0.5 h^{-1}.

(c) Find the residual glucose s at each D. Hint: set $\mu = D$ and solve for $f = s/(s + K_s)$; then $s = K_s f/(1 - f)$.

(d) Compute the biomass yield $Y_{sx} = D/v_1$. How does it change with D, and why?

(e) What is X_A at D = 0? What would the model predict with $k_3 = 0$, and why is that physically unrealistic?

Check your work: at D = 0.3 h^{-1} you should get $X_A = 0.148$ and s ≈ 10 mg L^{-1}.

Based on VNL eqs. (7.32a)–(7.32c) and Example 7.5

Solution 1 — Williams model

Steady state from the X_G balance; then the μ equation gives s

(a) X_G balance at steady state (μ = D, $X_G \neq 0$)

$$0 = \gamma_{22} k_2 X_A X_G - k_3 X_G - D X_G \;\Rightarrow\; X_A = \frac{D + k_3}{\gamma_{22} k_2} = \frac{D + 0.1}{2.7}$$

(c) Solve μ = D for the saturation f

$$f = \frac{s}{s + K_s} = \frac{D + (1 - \gamma_{22}) k_2 X_A X_G}{\gamma_{11} k_1}, \qquad s = K_s \frac{f}{1 - f}$$

(e) Resting cells

$$X_A(D = 0) = \frac{k_3}{\gamma_{22} k_2} = 0.037$$

With $k_3 = 0$ the model gives $X_A = 0$ at D = 0: resting cells would have no ribosomes on stand-by and could never restart growth. The G → A turnover is what keeps some A.

(b)–(d) Results

D (h^{-1})	X_A	X_G	f	s (mg L^{-1})	v_1	Y_{sx}
0.1	0.074	0.926	0.121	2.7	0.241	0.41
0.3	0.148	0.852	0.338	10.2	0.676	0.44
0.5	0.222	0.778	0.552	24.6	1.104	0.45

v_1 in g S (g DW h)$^{-1}$; Y_{sx} in g DW (g S)$^{-1}$

Reading the numbers

- X_A rises linearly with D — the ribosome trend of Fig. 7.11.
- Y_{sx} falls at low D: the loss term $(1 - \gamma_{22})v_2$ is a larger share of v_1 when growth is slow, so the model mimics maintenance without a maintenance term.
- Loss share $(1 - \gamma_{22})v_2/(\gamma_{11}v_1)$: 17 % at D = 0.1 but only 9 % at D = 0.5.

Check: at D = 0.5, μ = 0.5·1.104 − 0.1·3.0·0.222·0.778 = 0.552 − 0.052 = 0.500 h^{-1} ✓

Solution 1 explained — the plan and part (a)

Three unknowns (X_A, X_G, s): solve them in an order where each step has just one new unknown

Order of attack

1. **X_G balance → X_A, X_G**
   v_2 and v_3 contain no glucose, so this works before we know s
2. **$\mu = D$ → f**
   only equation containing glucose; linear in f
3. **f → s**
   undo the definition: $s = K_s\, f/(1 - f)$
4. **$v_1 = k_1 f$, $Y_{sx} = D/v_1$**
   plug-in results for part (d)

(a) Deriving X_A, step by step

1. Write the X_G balance at steady state (d/dt = 0) with $\mu = D$:
$$0 = \gamma_{22} v_2 - v_3 - D\,X_G$$
2. Insert the rate expressions $v_2 = k_2 X_A X_G$ and $v_3 = k_3 X_G$:
$$0 = \gamma_{22} k_2 X_A X_G - k_3 X_G - D\,X_G$$
3. Every term contains X_G. Divide it out ($X_G \neq 0$):
$$0 = \gamma_{22} k_2 X_A - k_3 - D$$
4. Solve for X_A, insert numbers, then $X_G = 1 - X_A$:
$$X_A = \frac{D + k_3}{\gamma_{22} k_2} = \frac{D + 0.10}{0.9 \times 3.0} = \frac{D + 0.1}{2.7}$$

Glucose never appeared in (a): the cell composition at a given D is set by the A ⇄ G balance alone. Glucose enters only in step 2 of the plan.

Solution 1 explained — part (c): find f, then s

μ = D is the only equation with glucose in it, and glucose appears only through f

(c) The algebra

1. Write μ = D with $v_1 = k_1 f$ and $v_2 = k_2 X_A X_G$:

$$\gamma_{11} k_1 f - (1 - \gamma_{22}) k_2 X_A X_G = D$$

2. X_A and X_G are known from (a), so f is the only unknown — and it is linear:

$$f = \frac{D + (1 - \gamma_{22}) k_2 X_A X_G}{\gamma_{11} k_1}$$

3. Convert the saturation back into a glucose concentration:

$$s = K_s \frac{f}{1 - f}$$

Reading the numerator: glucose uptake has to pay for growth (D) plus the active material lost when A is turned into G at only 90 % efficiency (the loss term).

Sanity check: f must lie between 0 and 1. If f ≥ 1, no glucose level can supply it — D is above the wash-out rate.

Worked example: D = 0.3 h^{-1}

Quantity	Calculation	Result
X_A	(0.3 + 0.1) / 2.7	0.148
X_G	1 − 0.148	0.852
Loss term	0.1 × 3.0 × 0.148 × 0.852	0.0379
f	(0.3 + 0.0379) / (0.5 × 2.0)	0.338
s	0.020 × 0.338 / (1 − 0.338)	0.0102 g L^{-1}
v_1	2.0 × 0.338	0.676
Y_{sx}	0.3 / 0.676	0.44

s = 0.0102 g L^{-1} = 10.2 mg L^{-1}. Repeat with D = 0.1 and 0.5 h^{-1} to fill the solution table.

Check at D = 0.3: μ = 0.5 × 0.676 − 0.0379 = 0.338 − 0.038 = 0.300 h^{-1} ✓

Solution 1 explained — parts (d) and (e)

Why the yield drops at low D, and why k_3 keeps some active material in resting cells

(d) Yield

$$Y_{sx} = \frac{\text{biomass made}}{\text{glucose used}} = \frac{\mu}{v_1} = \frac{D}{v_1}$$

μ is g DW per g DW per h and v_1 is g S per g DW per h, so their ratio is g DW per g S.

D (h^{-1})	v_1	Y_{sx}	Loss share
0.1	0.241	0.41	17 %
0.3	0.676	0.44	11 %
0.5	1.104	0.45	9 %

Loss share = $(1 - \gamma_{22})v_2 / (\gamma_{11}v_1)$: the part of the A made from glucose that is lost when A is turned into G. At slow growth it is a bigger fraction of a small uptake, so yield falls — the model behaves as if it had a maintenance term, without having one.

(e) What happens as D → 0

$$X_A(D=0) = \frac{k_3}{\gamma_{22}k_2} = \frac{0.10}{2.7} = 0.037$$

- $X_A = (D + k_3)/(\gamma_{22}k_2)$ is a straight line in D: slope $1/(\gamma_{22}k_2)$, intercept $k_3/(\gamma_{22}k_2)$.
- At D = 0 only the G → A turnover (k_3) keeps any active material in the cell.
- With $k_3 = 0$ the line passes through the origin: a non-growing cell would hold no active machinery (e.g. no ribosomes) at all.

Why that is unrealistic: real cells keep a stand-by pool of ribosomes — RNA content does not extrapolate to zero at μ = 0 (Fig. 7.11). $k_3 > 0$ gives the model that intercept.

Loss share at D = 0.1: (0.1 × 3.0 × 0.074 × 0.926) / (0.5 × 0.241) = 0.0206 / 0.121 = 17 %.

Problem 2 — LAB model: steady states in closed form

Two-compartment model of Nielsen et al. (1991) for Lactococcus, glucose-limited chemostat

Given

Symbol	Value	Units
γ_{11}	0.50	g X_A (g S)$^{-1}$
γ_{22}	0.90	g X_G (g X_A)$^{-1}$
k_1	3.0	h^{-1}
k_2	0.95	h^{-1}
$K_{s,1}$	15	mg L^{-1}
$K_{s,2}$	5	mg L^{-1}

Model

$$v_i = k_i\, f_i\, X_\mathrm{A}, \qquad f_i = \frac{s}{s + K_{s,i}} \qquad (i = 1, 2)$$

$$\mu = \gamma_{11} v_1 - (1 - \gamma_{22}) v_2$$

$$\frac{dX_\mathrm{G}}{dt} = \gamma_{22} v_2 - \mu X_\mathrm{G}, \qquad X_\mathrm{G} = 1 - X_\mathrm{A}$$

Tasks

(a) Show that at steady state $X_A = D / (D + \gamma_{22} k_2 f_2)$ and $D = \gamma_{11} k_1 f_1 - k_2 f_2$.
Hint: use the X_G balance first, then substitute into $\mu = D$.

(b) Tabulate D, X_A and $Y_{sx} = \mu / v_1$ for s = 20, 50, 200 and 1000 mg L^{-1}.

(c) Find the maximum dilution rate ($s \to \infty$) and the largest possible X_A.

(d) Below what glucose concentration s_{min} is no steady state possible? Derive s_{min} in closed form and explain its meaning.

Check your work: at s = 50 mg L^{-1} you should get D = 0.290 h^{-1} and X_A = 0.272.

Based on VNL Example 7.6, eqs. (1)–(7); parameters illustrative

Solution 2 — LAB model

Choose s, compute D: the steady state is explicit

(a) X_G balance, then μ = D

$$\gamma_{22}k_2f_2X_\mathrm{A} = D(1-X_\mathrm{A}) \;\Rightarrow\; X_\mathrm{A} = \frac{D}{D+\gamma_{22}k_2f_2}$$

$$\big(\gamma_{11}k_1f_1 - (1-\gamma_{22})k_2f_2\big)X_\mathrm{A} = D \;\Rightarrow\; D = \gamma_{11}k_1f_1 - k_2f_2$$

(c) Upper limit, s → ∞ ($f_1 = f_2 = 1$)

$$D_\mathrm{max} = \gamma_{11}k_1 - k_2 = 0.55\ \mathrm{h}^{-1}, \qquad X_\mathrm{A,max} = \frac{\gamma_{11}k_1 - k_2}{\gamma_{11}k_1 - (1-\gamma_{22})k_2} = 0.391$$

(d) Threshold: D = 0

$$s_\mathrm{min} = \frac{k_2K_{s,1} - \gamma_{11}k_1K_{s,2}}{\gamma_{11}k_1 - k_2} = \frac{0.95\cdot 15 - 1.5\cdot 5}{0.55} = 12.3\ \mathrm{mg\,L}^{-1}$$

(b) Steady states

s (mg L^{-1})	f_1	f_2	D (h^{-1})	X_A	Y_{sx}
20	0.571	0.800	0.097	0.124	0.456
50	0.769	0.909	0.290	0.272	0.463
200	0.930	0.976	0.469	0.360	0.467
1000	0.985	0.995	0.533	0.385	0.468

Reading the numbers

- D_{max} = 0.55 h^{-1} is the wash-out dilution rate quoted for L. cremoris in VNL.
- Below s_{min}, A is converted to G faster than it is made; $X_A \to 0$ as $D \to 0$, so resting cells lose their PSS — the same weakness as Williams with $k_3 = 0$.
- $Y_{sx} = \gamma_{11} - (1-\gamma_{22})(k_2/k_1)(f_2/f_1)$ rises with s because $K_{s,1} > K_{s,2}$.

Yield check at s = 50 mg L^{-1}: $Y_{sx} = 0.5 - 0.1\cdot(0.95/3)\cdot(0.909/0.769) = 0.463$ ✓

Solution 2 explained — what f_1 and f_2 are

Two reactions both need glucose, but they saturate at different glucose levels

Two reactions, two saturation factors

1 **Reaction 1: glucose → A**

makes active material

$$v_1 = k_1 f_1 X_A, \quad f_1 = \frac{s}{s + K_{s,1}}, \quad K_{s,1} = 15 \text{ mg L}^{-1}$$

2 **Reaction 2: A → G**

turns active into structural/genetic material

$$v_2 = k_2 f_2 X_A, \quad f_2 = \frac{s}{s + K_{s,2}}, \quad K_{s,2} = 5 \text{ mg L}^{-1}$$

- f_1 and f_2 are each reaction's own 'how busy' number, computed from the **same** glucose s.
- Both rates are proportional to X_A: A is the machinery that does both jobs.

f_1 and f_2 side by side

s (mg L^{-1})	f_1	f_2
5	0.250	0.500
15	0.500	0.750
20	0.571	0.800
50	0.769	0.909
200	0.930	0.976
1000	0.985	0.995

Why it matters

- $K_{s,2} < K_{s,1}$, so $f_2 > f_1$ at every s: reaction 2 (draining A) saturates first.
- At low s, A is drained nearly at full speed but made slowly — this creates the threshold s_{min} in part (d).

Compare with Williams (Problem 1): there only reaction 1 depended on glucose, so there was a single f. Here both do, so we need f_1 and f_2.

Solution 2 explained — pick s, get D

Running the problem backwards turns a quadratic into plug-in formulas

The strategy

Usual way (D → s): set D, then solve

$$D = 1.5\,\frac{s}{s+15} - 0.95\,\frac{s}{s+5}$$

for s. Clearing the fractions gives a quadratic — doable, but messy.

Backwards way (s → D):

choose s → f_1, f_2 → D → X_A → Y_{sx}

Every s you pick gives a genuine steady state — we are just labelling the steady states by s instead of by D. The table in the solution is built this way.

(a) Deriving the two formulas

1. X_G balance at steady state, with $\mu = D$, $X_G = 1 - X_A$, $v_2 = k_2 f_2 X_A$:
$$\gamma_{22} k_2 f_2 X_A = D\,(1 - X_A)$$

2. Collect the X_A terms:
$$X_A\,(D + \gamma_{22} k_2 f_2) = D \;\Rightarrow\; X_A = \frac{D}{D + \gamma_{22} k_2 f_2}$$

3. Now $\mu = D$, using $v_1 = k_1 f_1 X_A$ and $v_2 = k_2 f_2 X_A$:
$$\left[\gamma_{11} k_1 f_1 - (1 - \gamma_{22}) k_2 f_2\right] X_A = D$$

4. Substitute X_A from step 2, cancel D, multiply out:
$$\gamma_{11} k_1 f_1 - (1 - \gamma_{22}) k_2 f_2 = D + \gamma_{22} k_2 f_2$$

5. The two γ_{22} terms cancel:
$$D = \gamma_{11} k_1 f_1 - k_2 f_2$$

Reading it: growth rate = rate A is made − rate A is converted to G (both per unit A).

Step 4: $[\gamma_{11}k_1f_1 - (1-\gamma_{22})k_2f_2]\cdot D/(D + \gamma_{22}k_2f_2) = D$; divide by D, multiply by $(D + \gamma_{22}k_2f_2)$. Step 5: $-(1-\gamma_{22})k_2f_2 - \gamma_{22}k_2f_2 = -k_2f_2$.

Solution 2 explained — numbers, limits and the threshold

Worked example at s = 50 mg L^{-1}, then the two extremes: lots of glucose and too little

(b) Worked example: s = 50 mg L^{-1}

Quantity	Calculation	Result
f_1	50 / (50 + 15)	0.769
f_2	50 / (50 + 5)	0.909
D	0.5 × 3.0 × 0.769 − 0.95 × 0.909	0.290 h^{-1}
X_A	0.290 / (0.290 + 0.9 × 0.95 × 0.909)	0.272
v_1	3.0 × 0.769 × 0.272	0.627
Y_{sx}	0.290 / 0.627	0.463

D = 1.154 − 0.864 = 0.290 h^{-1}. Repeat for s = 20, 200 and 1000 mg L^{-1} to fill the table.

(c) Plenty of glucose: s → ∞, so $f_1 = f_2 = 1$

$$D_{\max} = \gamma_{11}k_1 - k_2 = 1.5 - 0.95 = 0.55 \text{ h}^{-1}$$

$$X_{A,\max} = \frac{D_{\max}}{D_{\max} + \gamma_{22}k_2} = \frac{0.55}{0.55 + 0.855} = 0.391$$

No feed rate can make cells grow faster than 0.55 h^{-1} — the wash-out limit used in Problem 3.

(d) Too little glucose: set D = 0

Growth stops when A is converted exactly as fast as it is made:

$$\gamma_{11}k_1\frac{s}{s+K_{s,1}} = k_2\frac{s}{s+K_{s,2}} \Rightarrow \gamma_{11}k_1(s+K_{s,2}) = k_2(s+K_{s,1})$$

$$s_{\min} = \frac{k_2K_{s,1} - \gamma_{11}k_1K_{s,2}}{\gamma_{11}k_1 - k_2} = \frac{14.25 - 7.5}{0.55} = 12.3 \text{ mg L}^{-1}$$

Below 12.3 mg L^{-1} the formula gives D < 0: no steady state. As D → 0, s → 12.3 mg L^{-1}, not 0 as in Monod.

In (d) we cancel one factor s on both sides (s > 0), then cross-multiply the denominators and solve the resulting linear equation for s.

Problem 3 — Wash-out: the culture remembers

Same LAB culture as Problem 2; two chemostats are shifted to $D = 0.99\ h^{-1}$ at $t = 0$ (above D_{max})

Given

Before the shift	Culture A	Culture B
D_0 (h^{-1})	0.10	0.50
s_0 (mg L^{-1})	20.3	337
X_{A0}	0.127	0.372

After the shift glucose rises within minutes to far above $K_{s,1}$ and $K_{s,2}$, so take $f_1 = f_2 = 1$ for $t > 0$.

Parameters as in Problem 2.

Useful result (logistic equation)

$$\frac{dX}{dt} = X(a - cX) \;\Rightarrow\; X(t) = \frac{X_\infty X_0\, e^{at}}{X_\infty + X_0\,(e^{at} - 1)}, \qquad X_\infty = \frac{a}{c}$$

Tasks

(a) Show that for $t > 0$, $\mu = cX_A$ and $dX_A/dt = X_A(a - cX_A)$, with $a = \gamma_{11}k_1 - k_2$ and $c = \gamma_{11}k_1 - (1 - \gamma_{22})k_2$. Evaluate a and c.

(b) Compute $\mu(0)$ and the initial wash-out rate $d \ln x/dt = \mu - D$ for each culture. What would a Monod model predict?

(c) Show that $\ln(x/x_0) = (a - D)t - \ln[X_A(t)/X_{A0}]$.

(d) Compute x/x_0 at $t = 1$, 2 and 3 h for both cultures, plot on a log axis, and compare with Fig. 7.13.

Check your work: $a = 0.55\ h^{-1}$ and $c = 1.405\ h^{-1}$; culture A starts with $\mu(0) = 0.18\ h^{-1}$.

Quantitative version of VNL Fig. 7.13 (Nielsen et al. 1991b). X_{A0} values are the Problem 2 steady states at D_0

Solution 3 — Wash-out

$a = 0.55\ h^{-1}$, $c = 1.405\ h^{-1}$, $X_\infty = a/c = 0.391$

(a), (c) Derivation

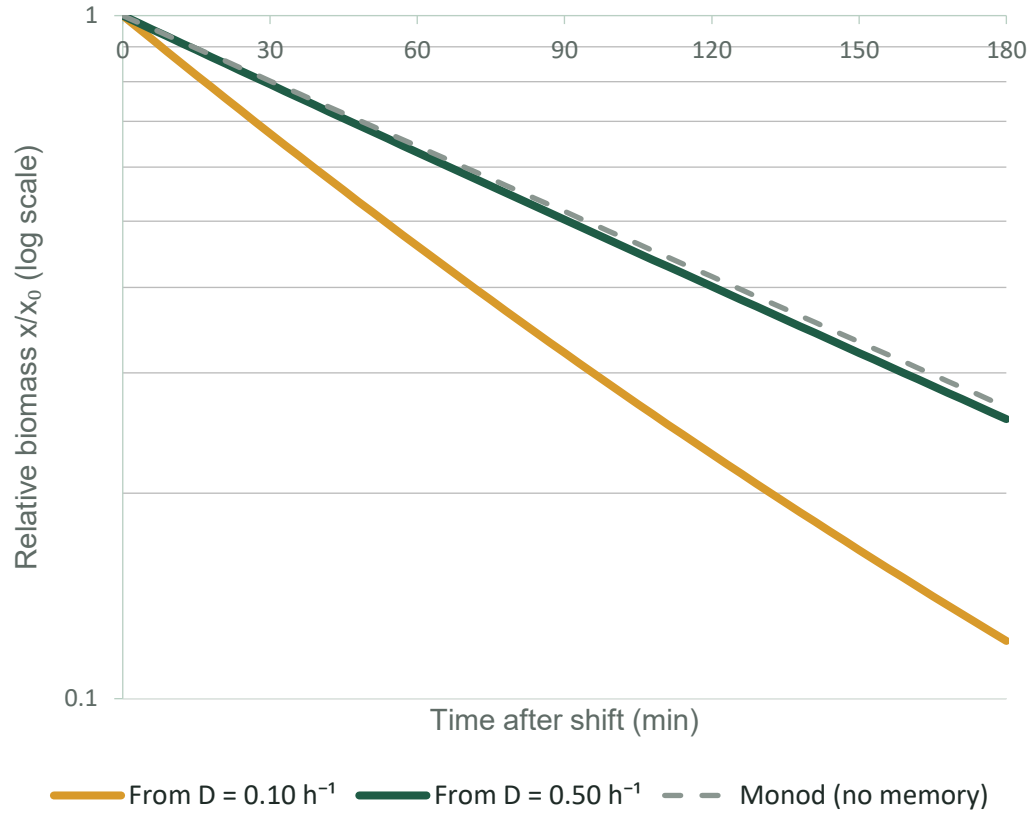
$$\frac{dX_A}{dt} = \gamma_{11} k_1 X_A - k_2 X_A - c X_A^2 = X_A (a - c X_A)$$

$$\frac{d \ln x}{dt} = c X_A - D, \quad \frac{d \ln X_A}{dt} = a - c X_A \Rightarrow \ln \frac{x}{x_0} = (a - D)\, t - \ln \frac{X_A}{X_{A0}}$$

(b), (d) Results

Culture	μ(0)	μ(0) – D	x/x_0 1 h	2 h	3 h
A (D_0 = 0.10)	0.179	−0.81	0.460	0.228	0.122
B (D_0 = 0.50)	0.523	−0.47	0.631	0.401	0.257
Monod	0.55	−0.44	0.644	0.415	0.267

Rates in h^{-1}. Culture A needs ≈ 5.3 h to build X_A to 90 % of X_∞ — it is washed out long before it adapts.



Compare Fig. 7.13 at 180 min: ≈ 0.15 (from 0.10 h^{-1}) vs ≈ 0.2 (from 0.50 h^{-1}). Both curves eventually become parallel with slope a − D = −0.44 h^{-1}

Solution 3 explained — why the culture ‘remembers’

After the shift, growth is limited by how much active machinery X_A the cells already have

The story in three steps

1 Before the shift

Each culture's X_A was set by its old dilution rate (Problem 2): slow culture A has $X_{A0} = 0.127$, fast culture B has 0.372.

2 Right after the shift

Glucose floods in, so $f_1 = f_2 = 1$. Glucose no longer limits growth — only X_A does: $\mu = cX_A$.

3 Consequence

A starts at $\mu = 0.18\ h^{-1}$, B at $0.52\ h^{-1}$. Both are below $D = 0.99\ h^{-1}$, so both wash out, but A faster — until it has built up X_A.

Monod has no memory: μ jumps straight to $\mu_{max} = 0.55\ h^{-1}$ whatever the history.

(a) Two equations once glucose is in excess

1 With $f_1 = f_2 = 1$: $v_1 = k_1X_A$, $v_2 = k_2X_A$, so

$$\mu = \gamma_{11}k_1X_A - (1-\gamma_{22})k_2X_A = c\,X_A$$
$$c = 1.5 - 0.1 \times 0.95 = 1.405\ \mathrm{h}^{-1}$$

2 Balance for A: made by reaction 1, used by reaction 2, diluted by growth:

$$\frac{dX_A}{dt} = \gamma_{11}v_1 - v_2 - \mu X_A = \underbrace{(\gamma_{11}k_1 - k_2)}_{a=0.55} X_A - cX_A^2$$

3 This is the logistic equation: X_A climbs from X_{A0} towards

$$X_\infty = a/c = 0.391 \quad (= X_{A,\max} \text{ of Problem 2})$$

Why logistic? A makes more A (aX_A: the more machinery, the faster it is built), but growth dilutes it (cX_A^2). The two balance at X_∞: the fully adapted composition.

The A balance is the mirror image of the G balance ($X_A + X_G = 1$). Adding them gives $\gamma_{11}v_1 - (1-\gamma_{22})v_2 - \mu = 0$, which is just the definition of μ.

Solution 3 explained — the log trick and the numbers

Adding two log-derivatives makes the unknown term cancel

(c) Deriving ln(x/x₀)

1. Biomass in the chemostat (sterile feed), divided by x:

$$\frac{d\ln x}{dt} = \mu - D = cX_\mathrm{A} - D$$

2. The X_A equation, divided by X_A:

$$\frac{d\ln X_\mathrm{A}}{dt} = a - cX_\mathrm{A}$$

3. Add them — the awkward cX_A term cancels:

$$\frac{d}{dt}\left[\ln x + \ln X_\mathrm{A}\right] = a - D$$

4. Integrate from 0 to t:

$$\ln\frac{x}{x_0} = \underbrace{(a-D)\,t}_{\text{adapted culture}} - \underbrace{\ln\frac{X_\mathrm{A}(t)}{X_\mathrm{A0}}}_{\text{memory term}}$$

(b), (d) Worked example at t = 1 h

Quantity	Culture A	Culture B
X_{A0}	0.127	0.372
$\mu(0) = cX_{A0}$	0.179 h^{-1}	0.523 h^{-1}
$\mu(0) - D$	−0.81 h^{-1}	−0.47 h^{-1}
X_A(1 h), logistic	0.178	0.380
Memory term	ln(0.178/0.127) = 0.337	ln(0.380/0.372) = 0.021
$\ln(x/x_0)$	−0.44 − 0.337 = −0.777	−0.44 − 0.021 = −0.461
x/x_0	0.460	0.631

(a − D)t = (0.55 − 0.99)(1) = −0.44. Monod (no memory term): $x/x_0 = e^{-0.44}$ = 0.644. B is already almost adapted ($X_{A0} \approx X_\infty$), so it tracks Monod; A pays a large memory penalty.

Logistic, culture A: $X_A(1) = 0.391 \times 0.127 \times e^{0.55} / (0.391 + 0.127 \times (e^{0.55} - 1)) = 0.0861 / 0.484 = 0.178$, with $e^{0.55} = 1.733$.

Problem 4 — Cybernetic model: who gets the resources?

Batch growth on glucose (i = 1) + xylose (i = 2) following the Kompala model, eqs. (7.33)–(7.38)

Given

Parameter	Glucose	Xylose
k_i (h^{-1}, per unit enzyme)	2.0	1.0
$K_i = K_i^E$ (g L^{-1})	0.01	0.05
γ_i (g DW g^{-1})	0.50	0.45
$k_{E,i}$ (h^{-1})	0.5	0.5
$k_{deg,i}$; $k_{con,i}$ (h^{-1})	0.05 ; 0.001	0.05 ; 0.001
State: s_i (g L^{-1})	0.5	1.0
State: $X_{E,i}$ (relative)	0.80	0.02

Model

$$r_i^* = k_i \frac{s_i}{s_i + K_i} X_{E,i}, \quad u_i = \frac{r_i^*}{\sum_j r_j^*}, \quad w_i = \frac{r_i^*}{\max_j r_j^*}$$

$$\frac{dX_{E,i}}{dt} = k_{E,i} \frac{s_i}{s_i + K_i^E} u_i - k_{\mathrm{deg},i} X_{E,i} + k_{\mathrm{con},i} - \mu X_{E,i}$$

Tasks

(a) Compute r_1^*, r_2^*, u_1, u_2, w_1, w_2.

(b) Find $\mu = \Sigma \gamma_i r_i^* w_i$ and the uptake rate of each sugar. Is xylose being used?

(c) Compute $dX_{E,1}/dt$ and $dX_{E,2}/dt$. Why do both enzyme levels fall?

(d) Holding $X_{E,i}$ fixed, down to what glucose concentration does glucose stay preferred ($u_1 > u_2$)?

(e) When glucose runs out, $X_{E,2} = 0.02$ and $s_2 \approx 1$ g L^{-1}. Estimate the diauxic lag: the time for μ to reach half its final value. Hint: early on, $\mu X_{E,2}$ and $k_{deg} X_{E,2}$ are small.

Check your work: r_1^* = 1.57 g S (g DW h)$^{-1}$, so u_1 is very close to 1.

Based on VNL Sect. 7.4.2; Kompala et al. (1986); Turner & Ramkrishna (1988). Enzyme levels are relative (fully induced ≈ 1)

Solution 4 — Cybernetic model

Glucose takes almost everything until it is essentially gone

(a)–(c) At the given state

Quantity	Glucose	Xylose
r_i^* (g S g^{-1} DW h^{-1})	1.569	0.0190
u_i	0.988	0.012
w_i	1.000	0.012
Uptake $r_i^* w_i$	1.569	0.00023
$dX_{E,i}/dt$ (h^{-1})	−0.182	−0.010

(b) $\mu = 0.5 \cdot 1.569 + 0.45 \cdot 0.00023 =$ **0.784** h^{-1}. Xylose uptake is ~7000× smaller — it is effectively not used.

(c) Fast growth dilutes both enzymes: E_1 relaxes toward (0.484 + 0.001)/(0.05 + 0.784) = 0.58; E_2 keeps shrinking, so the longer the culture grows on glucose, the longer the later lag.

(d) Switch point: $r_1^* = r_2^*$

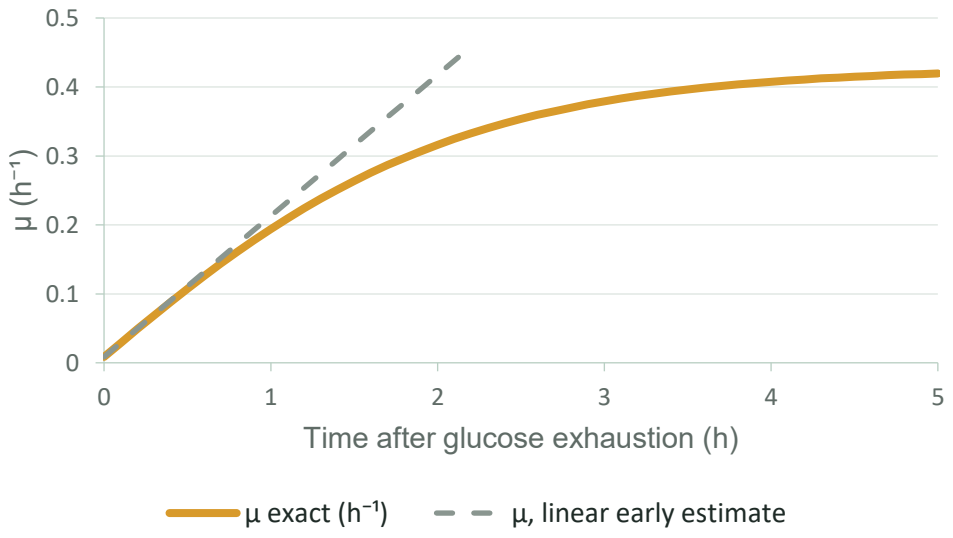
$$2.0 \cdot 0.8 \frac{s_1}{s_1 + 0.01} = 0.0190 \;\Rightarrow\; s_1 \approx 0.12 \text{ mg L}^{-1}$$

Glucose stays preferred until it is essentially exhausted → a sharp diauxic switch.

(e) Diauxic lag on xylose ($u_2 = w_2 = 1$)

$$\frac{dX_{E,2}}{dt} = A - BX_{E,2} - CX_{E,2}^2, \quad A = 0.477,\ B = 0.05,\ C = 0.429 \text{ h}^{-1}$$

Final state: $X_{E,2} = 0.998$, $\mu_\infty = CX_{E,2} = 0.428$ h^{-1}. Early on $X_{E,2} \approx 0.02 + At$, so $\mu = \mu_\infty/2$ when $X_{E,2} \approx 0.499$: **t ≈ 0.48/0.477 ≈ 1.0 h** (exact 1.13 h; 90 % at 3.2 h).



A = $k_{E,2}f_2 + k_{con,2}$; B = $k_{deg,2}$; C = $\gamma_2 k_2 f_2$; $f_2 = 1/(1 + 0.05) = 0.952$. Exact curve by numerical integration

Solution 4 explained — the cybernetic idea in plain words

The cell is treated as an investor with limited resources, choosing which sugar pathway to back

r_i^* Potential return

How fast pathway i could take up its sugar right now, given the sugar level and how much enzyme $X_{E,i}$ the cell already has.

$$r_i^* = k_i \frac{s_i}{s_i + K_i} X_{E,i}$$

u_i Investment: which enzyme to make

Share of enzyme-making capacity spent on enzyme i. Proportional to return; $u_1 + u_2 = 1$ (matching law).

$$u_i = \frac{r_i^*}{\sum_j r_j^*}$$

w_i Activity: which enzyme to use

How much existing enzyme i may work. Best pathway runs fully (w = 1); others are throttled by their ratio to the best.

$$w_i = \frac{r_i^*}{\max_j r_j^*}$$

Putting it together

$$\mu = \sum_i \gamma_i \, r_i^* \, w_i$$

yield × actual uptake rate, summed over both sugars

$$\frac{dX_{E,i}}{dt} = \underbrace{k_{E,i} \frac{s_i}{s_i + K_i^E} u_i}_{\text{synthesis}} - \underbrace{k_{\text{deg},i} X_{E,i}}_{\text{decay}} + \underbrace{k_{\text{con},i}}_{\text{constitutive}} - \underbrace{\mu X_{E,i}}_{\text{dilution by growth}}$$

Enzyme i is made in proportion to u_i, decays slowly, has a small always-on (constitutive) production, and is diluted as the cells grow and divide.

u controls the long term (induction: which enzyme gets made); w controls the short term (inhibition: which enzyme gets used). Here $K_i^E = K_i$.

Solution 4 explained — parts (a) to (d) with numbers

Glucose wins on every count, and xylose is penalised twice

(a)–(c) Line by line

Quantity	Glucose (i = 1)	Xylose (i = 2)
$f_i = s_i/(s_i + K_i)$	0.5/0.51 = 0.980	1.0/1.05 = 0.952
$r_i^* = k_i f_i X_{E,i}$	2.0 × 0.980 × 0.80 = 1.569	1.0 × 0.952 × 0.02 = 0.0190
$u_i = r_i^*/(r_1^* + r_2^*)$	1.569/1.588 = 0.988	0.0190/1.588 = 0.012
$w_i = r_i^*/\max r^*$	1.569/1.569 = 1	0.0190/1.569 = 0.012
Uptake $r_i^* w_i$	1.569	0.0190 × 0.012 = 0.00023
Synthesis $k_E f_i u_i$	0.5 × 0.980 × 0.988 = 0.484	0.5 × 0.952 × 0.012 = 0.0057
Decay $-k_{deg} X_{E,i}$	−0.040	−0.001
Constitutive $+ k_{con}$	+0.001	+0.001
Dilution $-\mu X_{E,i}$	−0.784 × 0.80 = −0.627	−0.784 × 0.02 = −0.016
$dX_{E,i}/dt$ (sum)	**−0.182 h^{-1}**	**−0.010 h^{-1}**

What the numbers say

- **(b)** $\mu = 0.5 \times 1.569 + 0.45 \times 0.00023 = 0.784\ h^{-1}$.
- Xylose is hit twice: little enzyme makes r_2^* small, then w_2 throttles it again → ~7000× less uptake.
- **(c)** Both enzymes fall because dilution by fast growth (μX_E) outweighs synthesis.

(d) When does glucose lose?

$u_1 > u_2$ exactly when $r_1^* > r_2^*$ (same denominator). Set them equal:

$$1.6\,\frac{s_1}{s_1 + 0.01} = 0.0190 \;\Rightarrow\; s_1 = 0.12\ \text{mg}\,\text{L}^{-1}$$

Glucose stays preferred until it is essentially gone.

(d) algebra: $1.6\ s_1 = 0.0190\ (s_1 + 0.01) \Rightarrow s_1 = 0.01 \times 0.0190 / (1.6 - 0.0190) = 1.2 \times 10^{-4}$ g L^{-1}. Here $1.6 = k_1 X_{E,1} = 2.0 \times 0.8$.

Solution 4 explained — part (e): the diauxic lag

Once glucose is gone the cell must build xylose enzyme almost from scratch

Step by step

1. Glucose gone ⇒ $r_1^* = 0$, so $u_2 = w_2 = 1$. Growth is on xylose only:

$$\mu = \gamma_2 k_2 f_2 X_{E,2} = C\, X_{E,2}$$

2. Enzyme balance with $u_2 = 1$ (write X for $X_{E,2}$):

$$\frac{dX}{dt} = A - BX - CX^2, \quad A = k_{E,2} f_2 + k_{\text{con}} = 0.477,\ B = 0.05,\ C = 0.429$$

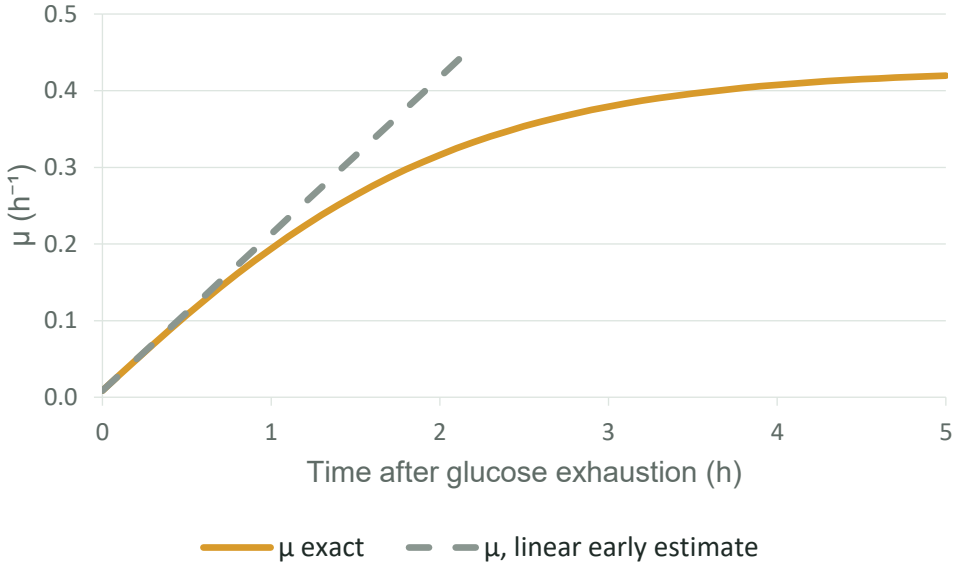
3. Final state: set dX/dt = 0 and solve the quadratic:

$$X_\infty = \frac{-B + \sqrt{B^2 + 4AC}}{2C} = 0.998, \quad \mu_\infty = C X_\infty = 0.428\ \text{h}^{-1}$$

4. Half of μ_∞ means X = 0.499 ($\mu \propto X$). Early on BX, $CX^2 \ll A$, so $X \approx 0.02 + At$:

$$t_{1/2} \approx \frac{0.499 - 0.02}{0.477} \approx 1.0\ \text{h}$$

Link to (c): the lag depends on how much xylose enzyme is left when glucose runs out (here 0.02). The longer the culture grew on glucose, the lower $X_{E,2}$, the longer the lag.



Exact answer: 1.13 h (90 % at 3.2 h). The estimate is a bit short because by X ≈ 0.5 the ignored terms BX + CX² ≈ 0.13 are about a quarter of A — decay and dilution have started to slow the build-up.

$f_2 = 1/(1 + 0.05) = 0.952$; $A = 0.5 \times 0.952 + 0.001$; $C = 0.45 \times 1.0 \times 0.952$. Exact curve by numerical integration of $dX/dt = A - BX - CX^2$.